

Predictive Model Plan – Geldium Delinquency Prediction

1. Model Logic (Generated with GenAI)

Purpose:

The model is designed to predict whether a customer is likely to become delinquent by estimating a probability based on their financial and payment history.

Simplified Model Logic (Pseudo-code):

Start

Load customer dataset

Select input features:

- Month_1
- Credit_Utilization
- Income
- Debt_to_Income_Ratio
- Credit_Score

Preprocess data:

- Handle missing values
- Validate data types
- Prepare features for training

Train Logistic Regression model

For each customer:

 Calculate probability of delinquency

If probability $\geq$ decision threshold:

 Predict High Risk

Else:

 Predict Low Risk

Generate risk score and prediction

End

The Logistic Regression model analyzes the selected customer features to estimate the probability of delinquency. Based on the calculated probability, it assigns each customer a risk score between 0 and 1 and classifies them as either Low Risk or High Risk, supporting credit risk assessment and lending decisions.

2. Justification for Model Choice

Logistic Regression was selected because it offers a strong balance between predictive performance, transparency, and implementation simplicity for binary classification problems such as credit delinquency prediction. Compared with more complex models such as Neural Networks, Logistic Regression produces explainable predictions, allowing business stakeholders, auditors, and regulators to understand how lending decisions are made. The model is computationally efficient, straightforward to implement, and widely used in financial risk assessment, making it well suited to Geldium's business needs where interpretability, regulatory compliance, and reliable decision-making are as important as predictive performance.

3. Evaluation Strategy

The predictive model will be evaluated using multiple evaluation metrics to provide a balanced assessment of its performance. Accuracy will measure the overall proportion of correctly classified customers, while the F1 Score will evaluate the balance between precision and recall, making it particularly useful if the dataset contains an imbalance between delinquent and non-delinquent customers. The AUC (Area Under the ROC Curve) will measure the model's ability to distinguish between high-risk and low-risk customers across different classification thresholds.

To promote fairness, model performance will be compared across different demographic groups to identify potential bias or significant differences in prediction outcomes. If disparities are detected, appropriate mitigation techniques, such as reviewing feature selection, balancing the training data, or refining the model, will be considered to support fair and responsible decision-making.

Ethically, the model should support human decision-making rather than replace it, and its predictions should be used responsibly while maintaining transparency and compliance with financial regulations. Model performance should also be monitored after deployment to ensure prediction quality remains consistent over time.

4. Appendix: GenAI Prompts Used

To assist in structuring and refining this predictive modelling plan, the following prompts were utilised:

Prompt for Model Logic: "Act as a data scientist. Write a simple conceptual workflow and pseudocode for a logistic regression model predicting credit delinquency. Use input features like Month_1 (Recent Payment History), Credit Utilisation, Income, Debt to Income Ratio, and Credit Score."

Prompt for Model Justification: "Write a 1-paragraph business justification for choosing logistic regression over neural networks for predicting credit risk at a financial institution. Focus on its transparency, interpretability, and regulatory compliance."

Prompt for Evaluation Strategy: "Outline an evaluation strategy for a credit risk model. Explain the use of metrics like Accuracy, F1 Score, and AUC. Include a section on how the model should be audited for fairness, bias, and ethical deployment."